

CANDIDATE
NAME

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INDEX NUMBER

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BIOLOGY

9648/03

Applications Paper and Planning Question
Paper 3

20 September 2016
Tuesday

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and PD group on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graph or rough working.
Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

All working for numerical answers must be shown.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

Calculators may be used

For Examiner's Use	
1	
2	
3	
4	
5	
Total	72

Answer **all** the questions.

1

(a) State **three** differences between a genomic DNA library and a cDNA library.

Genomic Library	cDNA Library
A collection of clones containing fragments of genomic DNA representing the entire genome of a species;	A collection of clones containing cDNA copies of all the mRNAs (expressed genes) in a cell type at a particular time;
Library size is larger;	Library size is smaller and more compact;
A Genomic library is derived from the DNA of the entire genome of an organism;	A cDNA library is derived from the mRNA isolated from a cell of the organism;
DNA sequences contained include exons, introns and regulatory sequences;	cDNA sequences contain only exons / lack intron and regulatory sequences;
Gene sequences are not intact because they are cut by restriction enzymes at restriction sites;	Gene sequences are intact because cDNA are reverse-transcribed from mRNA;
More difficult to find desired gene sequence;	Easier to find desired gene sequence;
Vectors include bacteriophage lambda, cosmid, bacteriophage P1, bacterial artificial chromosome and yeast artificial chromosome;	Vectors used include plasmid or bacteriophage lambda;

[3]

Many countries are trying to increase their production of bioethanol in order to fuel a range of vehicles. Most bioethanol is produced from the cellulose cell walls of plants. The cellulose synthase gene have been identified and sequenced.

In initial experiments to increase cellulose production, DNA containing the cellulose synthase gene were inserted into plasmid vectors and subsequently introduced into bacteria cells through transformation. Fig. 1.1 is a diagram of the pUC 19 plasmid map used in the experiments.

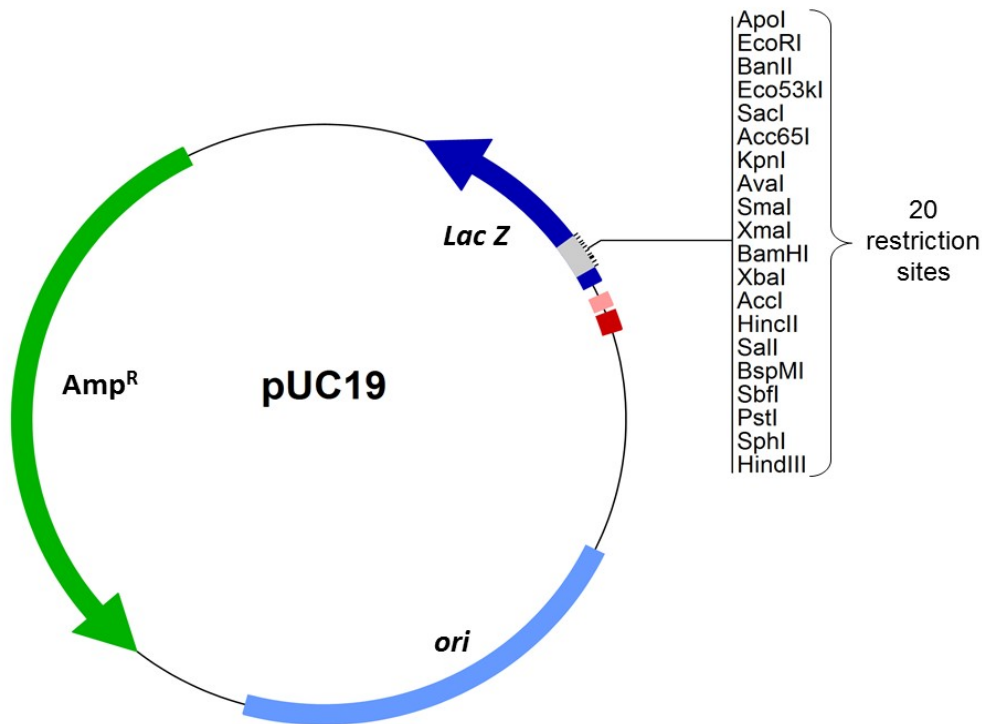


Fig. 1.1

(b) (i) State and explain **two** visible properties of the pUC 19 plasmid that allow it to be used as a DNA cloning vector.

- Contain **two selectable marker genes** (need to cite amp and LacZ);
- Expression of genes confer well-defined phenotype on the host cell, allowing for selection of transformed cells which have taken up vector or recombinant DNA molecules;
- contains **multiple cloning site / polylinker**;
- allow **20 different restriction enzymes** to cleave the plasmid DNA to insert some / many genes;
- have **origin of replication**;
- plasmid can be **replicated independently** of bacteria chromosome, to give **multiple copies** of the plasmid and gene of interest within **one bacterium**, ensures gene of interest can be passed onto daughter cells;

[max 4m]

[4]

(ii) Explain how successfully transformed bacterial cells containing recombinant plasmids can be selected for.

- Bacterial cells grown on **medium** containing **ampicillin** and **X-gal**;
- Bacterial cells that **survive** and **form white colonies** are successfully transformed cells containing recombinant plasmids;
- Bacterial cells that survive **contain ampicillin gene** which is **transcribed and translated** to produce **ampicillin resistance proteins**;
- Cells that contain recombinant plasmids will have where there is **insertional inactivation of lacZ gene**, **no functional β -galactosidase enzymes** produced through transcription and translation and **X-gal not hydrolysed** to form blue products;

[max 3m]

[3]

(c) The cellulose synthase gene obtained from a plant species has approximately 5000 base pairs (bp). However the length of the gene sequence inserted into bacterial cells was shortened to only approximately 3300bp.

Explain why the length of the gene inserted into bacterial cells was shortened.

- **Introns** from the gene sequence removed / gene **only** contain **exons**;
- **Lack of splicing mechanism** in bacterial cells;
- Translation of mRNA would produce a polypeptide chain with correct amino acid sequence;

[max 2m]

[2]

(d) Using bacteriophages to introduce the cellulose synthase gene into bacterial cells was thought to improve transformation efficiency. However, a lower than expected proportion of bacterial cells was found to contain the gene after bacterial transduction using bacteriophages.

Explain why using bacteriophages to introduce DNA into bacterial cells was not successful.

- Bacterial cells have **restriction enzymes / endonucleases**;
- **Cleave / hydrolyse the DNA** containing the gene of interest to **protect bacterial cells from viral / bacteriophage infections**;

[2]

[Total: 14]

2 Antithrombin (AT) is a protein required for the prevention of blood clots in the blood plasma. Patients with hereditary AT deficiency requires AT injection before they undergo any surgery to prevent the formation of blood clot in the veins.

ATryn, a recombinant AT (rAT) is the first medicine produced using genetically engineered animals. rAT is made from the mammary glands of goats that have been genetically modified (GM). The DNA construct introduced into goats comprises of the cDNA of human AT and regulatory elements of the goat beta casein gene, an essential milk protein produced in goat mammary glands. Microinjection was used to insert the DNA construct into a goat zygote which is then transplanted into the uterus of a female goat. GM progeny goat will produce large amounts of AT into its milk which are then isolated and purified.

(a) Explain why the DNA construct used is a recombinant DNA.

- **Comprises of DNA from 2 different species / sources: human and goat**

[1]

- (b) Suggest why the DNA construct used must include regulatory elements of the goat beta casein gene.

[2]

- Regulatory elements of the goat casein gene can be bounded by transcription factors present in (cells of) mammary gland
- To allow expression/ transcription and translation of the human AT gene only in the goat mammary gland

The effectiveness of rAT can be measured in terms of percentage activity compared with normal human blood plasma AT (hpAT). rAT was injected into four patients with AT deficiency (AT activity $\leq 60\%$) and their AT activities were measured every two days after the injection. Fig. 2.1 shows the results of the study.

Table 2.1 shows a comparison between the structure of rAT and hpAT.

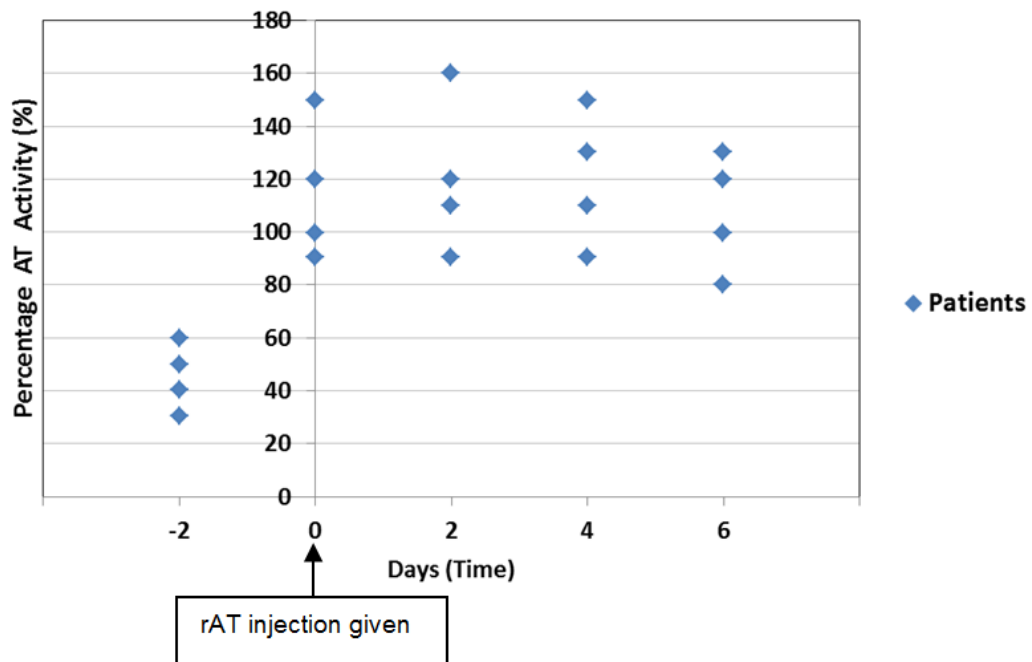


Fig. 2.1

	rAT	hpAT
Primary structure	Single chain of 432 amino acids	Single chain of 432 amino acids
Glycosylation	Glycosylation at 3 asparagine amino acid residues (Asn 96, 155, 192)	Glycosylation at 4 asparagine amino acid residues (Asn 96, 135, 155, 192)
Analysis of carbohydrate chains on the protein	Mainly N-acetyl neuraminic acid	Mainly N-glycolyl neuraminic acid

Table 2.1

(c) Using the information above,

- (i) describe the effect of injecting rAT on the patients .
- AT activity in all patients increased from a range of 30-60% AT activity before treatment to 80-160% AT after treatment

[1]

- (ii) describe a difference between the structures of rAT and hpAT and the effect of the difference on the patients.

Describe the difference (any 1)

- Glycosylation at 4 asparagine amino acid residues while rAT had glycosylation at 3 asparagine amino acid residues/ Asn 135 in rAT is not glycosylated
- rAT has mainly N-acetyl neuraminic acid attached while hpAT has mainly N-glycolyl-neuraminic acid

Effects on patients

rAT has different tertiary structure from hpAT → Patients have AT activity (up to 160% which is 60%) more than normal human AT

[2]

- (iii) suggest what resulted in the differences in the structures of rAT and hpAT.

- different enzymes carrying out glycosylation in goat and human
- N-glycolyl neuraminic acid not synthesised / found in goat cells and vice versa
- AVP

[1]

Polymerase chain reaction (PCR) and gel electrophoresis can be used to distinguish between a DNA construct containing both cDNA of human AT and regulatory elements and one which contains only the regulatory elements.

Fig. 2.2 shows the DNA construct used in forming rAT.



Fig. 2.2

- (d) (i) On Fig. 2.2, indicate with **5' → 3'** where the primers will bind in the PCR reaction.

5' and 3' represents the 5' and 3' ends of the primers respectively.

[1]

- 2 arrows either flanking the region containing both regulatory elements and cDNA or flanking/ on the cDNA of human AT

- (ii) Explain how gel electrophoresis can be used to distinguish between a DNA construct with both cDNA of human AT and regulatory elements and one which contains only regulatory elements.

- gel electrophoresis to separate the PCR products by **size** under a **direct current**

- DNA containing AT and regulatory elements are bigger/larger, hence travel slower across the gel / DNA band appear nearer to the cathode *accept converse* [2]

OR

- A (1296 bp) band will be observed for DNA containing AT and promoter while no bands observed for DNA containing only regulatory elements

(e) A new project, GenomeAsia 100K, aims to sequence the genomes of 100,000 Asians across Asia to tailor-made drugs for Asians, an approach known as precision medicine.

(i) State **one** benefit of precision medicine.

- greater drug efficacy
- prevents dangerous side effects
- AVP

[1]

(ii) While there are many benefits of precision medicine, some people are apprehensive that the project will unravel genetic differences in ethnicity “making some people a lesser man and others a better man”

Explain why the project may result in the ethical issue of “making some people a lesser man and others a better man”.

- Discrimination/labelling/stereotyping of certain ethnic group based on **genetic profile/genome**
- Idea of genetic influence on behaviour, character traits, susceptibility to diseases e.g. People will be tempted to draw connections between genes and propensity to violence, intelligence, creativity etc

[2]

[Total: 13]

3 In 2006, scientists Takahashi and Yamanaka stimulated mouse tail-tip cells, which are fully differentiated cells, to change back into stem cells in tissue culture. Such stem cells obtained from the de-differentiation of somatic cells are known as induced pluripotent stem cells (iPS cells). In the last decade, iPS cells have been generated from humans cells isolated from different tissues by introduction of various growth factors that affects the expression of many genes, such as the gene coding for human telomerase reverse transcriptase (hTERT).

(a) Explain how the expression of hTERT may help the cells regain stem cell properties.

- hTERT catalyses the formation of phosphodiester bonds between DNA nucleotides which are added to the 3' end of parental DNA strands (via complementary base pairing with telomerase RNA) during DNA replication;
- Increases length of telomeres/prevents telomeres from shortening → prevents telomeres from reaching critical length after few rounds of DNA replication;
- iPS cells are able to self-renew and divide by mitosis continuously/repeatedly (must have idea of many rounds of cell division);

[3]

(b) X-linked Severe Combined Immunodeficiency (SCID) is a rare congenital disorder characterised by improper development of immune cells which has been treated by gene therapy.

Explain why X-linked SCID can be treated by gene therapy.

- X-linked-SCID is a single gene disorder/idea of only one gene mutated OR sequence of the IL2RG gene is well characterised
- Introduction/ insertion of only one normal interleukin 2 receptor gamma (IL2RG) allele into target cells is sufficient to mask the phenotypic effect of recessive mutation;

OR

- Normal IL2RG dominant allele introduced into the targeted blood stem cell codes for sufficient common gamma chain of (several) cytokine receptors (via transcription and translation of the normal IL2RG allele);

[2]

The ability to generate iPS cells that have characteristics that are similar to embryonic stem cells has provided a promising alternative to the use of haematopoietic stem cells for gene therapy to treat X-linked SCID. Fig. 3.1 shows a possible process of using iPS cells for gene therapy.

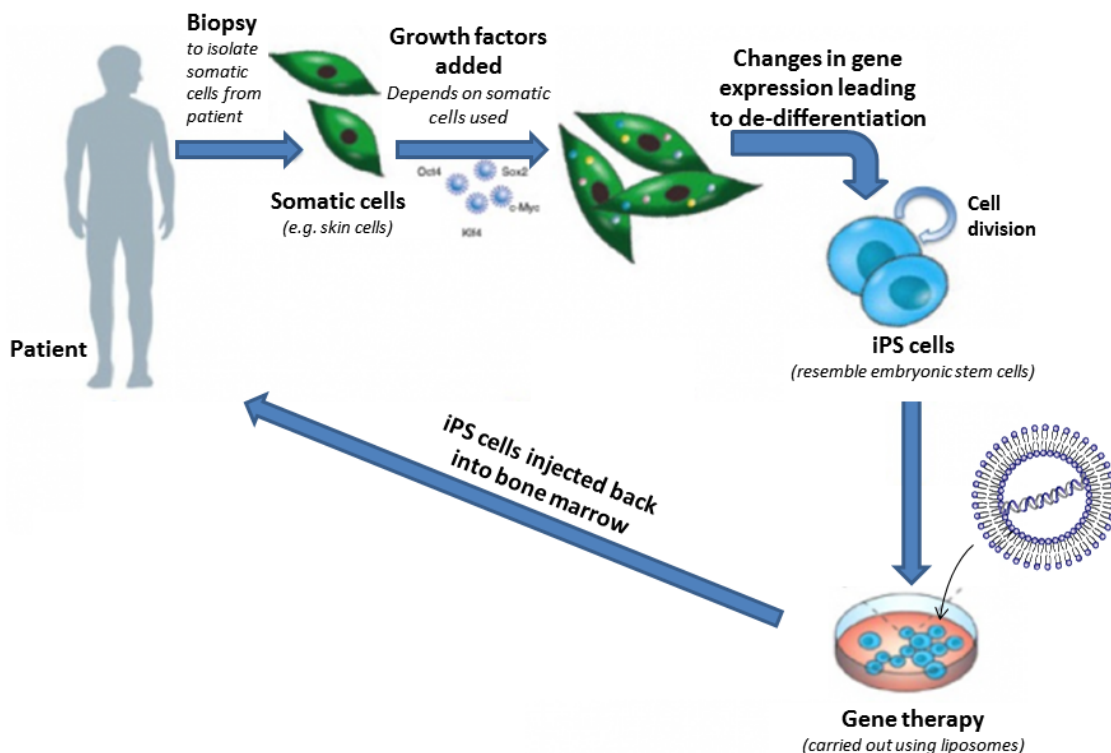


Fig. 3.1

- (c) Describe and explain **two** factors that prevent this method of gene therapy for X-linked SCID from becoming an effective treatment.

Difficulties in de-differentiating somatic cells to iPS cells

- Difficult to ensure that specialised/completely differentiated somatic cells become pluripotent (completely) as it is difficult to determine the growth factors to be used;
- Treatment may not be effective as the iPS cells obtained are unable to differentiate into blood stem cells in the bone marrow;

Use of liposome as vector

- Lower probability of liposomes binding to cell surface membrane of iPS cells and membrane fusion occurring to release DNA into target cells;
- therefore not all target cells receive normal IL2RG allele, some still express non-functional IL2RG protein;

OR

- Normal IL2RG allele not integrated into the iPS cell's DNA, unless retroviral vector is used;

[4]

- this leads to transient expression of normal functional proteins and multiple treatments are required;

N2011 Examiner's report: Candidates should take care not to suggest that genes die or do not live for very long. Also note that retroviral vectors may not be suitable for treatment of all genetic diseases as they may not be able to infect certain target cells

Gene therapy does not offer complete cure

- Difficult to control the expression of the normal IL2RG alleles
- Expression of proteins may be unstable – there may be too much or too little proteins being expressed → immune cells may have insufficient cytokine receptors/are abnormal, cannot carry functions;

- (d) Although the use of retroviral vectors for gene therapy has been highly successful in treating SCID, many patients who received treatment also developed immune responses against the virus and leukaemia.

Suggest another challenge of using viral vectors in the process shown in Fig. 3.1.

- Difficult to find viral vectors which have surface glycoproteins that are complementary in shape/can bind to the cell surface receptors of the target iPS cells;

[1]

- (e) (i) Discuss the arguments for and against the use of human embryonic stem cells for therapy.

- Against use of ES cells
 - o involves removal of inner cell mass from blastocyst → destruction of embryo which has the potential/ability to develop into a foetus/a human being → akin to murder/killing of a life for own benefit/to treat own disease;
 - o ES cells are able to divide continuously via mitosis → potential of developing tumours, causing more harm to the patient.
- For use of ES cells
 - o unclear whether ES cells are considered human life → (it does not have any interests to be protected) and thus should be used to benefit/treat/save patients with life-threatening diseases;
 - o ES cells are pluripotent → able to differentiate into any other cell types of the 3 germ layer → potential to treat diseases where harvesting of adult stem cells are difficult;

[2]

- (ii) Explain whether you agree or disagree that use of iPS cells resolve the issues discussed above.

- Agree → iPS cells were not harvested from embryos but were obtained from de-differentiation of somatic cells → no destruction of human life so it is acceptable;

OR

- Disagree → iPS cells have characteristics of ES cells and hence have the potential of developing into a develop into a foetus/a human being;

(Student must state a clear stand to gain mark)

[1]

[Total: 13]

4 Planning question

Diffusion of molecules can be observed using the Visking tubing.

The Visking (dialysis) tubing is a selectively permeable membrane that allows small molecules to diffuse through. The tubing can be tightly knotted at one end and a solution can be contained within the tubing. The filled tubing can then be placed in distilled water for the diffusion of the solute molecules out of the tubing into the distilled water. Fig. 4.1 shows the Visking tubing set-up as described.

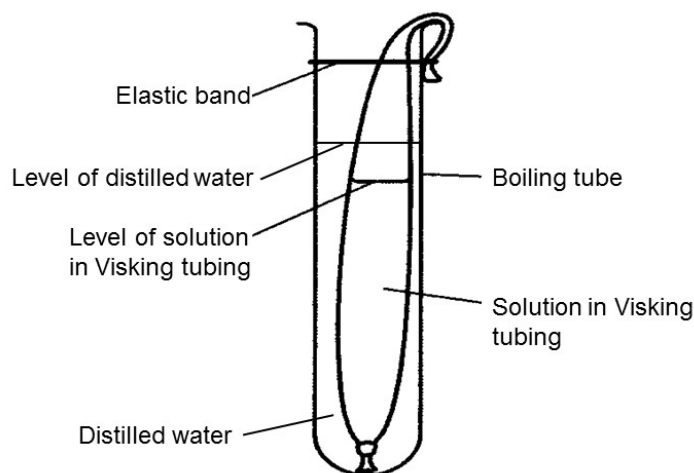


Fig. 4.1

The presence of solutes in distilled water surrounding the tubing can then be tested using quantitative / qualitative analysis.

Presence of glucose in solution can be tested using Benedict's test. The concentration of glucose in a solution can be estimated using a colour standard made with solutions with known glucose concentrations.

Using this information and your own knowledge, design an experiment to investigate the effect of temperature on the diffusion of glucose.

You must use:

- 100 cm³ of 10g / 100 cm³ glucose solution,
- 10 cm³ of 5g / 100 cm³ glucose solution for colour standard,
- Distilled water,
- Thermostatically-controlled water bath and thermometer,
- Benedict's solution,
- Boiling tubes,
- Stopwatch.

You may select from the following apparatus and chemicals:

- Normal laboratory glassware, e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods etc.,
- Syringes.

Your plan should:

- Have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12]

Theory of Explanation [0.5m each, 1m]

Effect of Temperature on diffusion of Glucose

- The higher the temperature, the higher the **kinetic energy** of glucose molecules;

Benedict's Test

- Producing brick red precipitate from reduction of copper (II) ions;

Hypothesis [0.5m each, 1m]

- The **higher the temperature**, the **faster the diffusion of glucose / more glucose diffuse** into the surrounding distilled water / out of the Visking tubing;
- The greater the concentration of glucose in distilled water, the **greater the amount of brick-red precipitate** produced in the reducing sugar test;

Step-by-step experimental procedure [0.5m each, max 5m]

Making of glucose standard solutions [0.5m each, max 1m]

- Using serial dilution / simple dilution, concentration stated 0.1 g – 5 g per 100 cm³ (any reasonable range);
- Description of dilution process / drawing of dilution table;

Diffusion of glucose at different temperatures [0.5m each, max 2.5m]

- Tie one end of the Visking tubing;
- Use syringe to add 10 cm³ of 10 g per 100 cm³ glucose solution into Visking tubing (accept any reasonable volume of solution);
- Use syringe to add 20cm³ of distilled water into a boiling tube (accept any reasonable volume of solution);
- Place the boiling tube into the **thermostatically controlled water bath** set at 30°C;
- Time 1 minute using a stopwatch for **equilibration**;
- Measure the temperature of the water bath water using a **thermometer**;

- Place Visking tubing into the boiling tube and time for **2 minutes** using **stopwatch** (accept any fixed period of time);
- After 2 minutes, remove the boiling tube from the water bath and remove the Visking tubing from the boiling tube;
- Repeat the experiment at different temperatures, (Accept any temperature range from 30°C to 100°C, five stated temperatures required, reject any temperatures below 25°C);
- Dependent variable: Rate of glucose diffusion;
- Independent variable: Temperature;

Reducing sugar test [0.5m each, max 1.5m]

- Add 2cm³ of solution from one boiling tube and 1cm³ of Benedict's solution into a test tube using 5cm³ syringe;
- Place test tubes into a thermostatic water bath set at 90°C / boiling water bath for 2 minutes (accept reasonable time);
- Repeat with the solutions from four boiling tubes and the glucose solutions of known concentration;
- **Compare the colour of the precipitates** from the solutions of the boiling tubes with the colour of the precipitates from the glucose solutions of known glucose concentration to **determine the glucose concentration of solutions from the boiling tubes**;

Reliability and Reproducibility [0.5m each, max 1m]

- Carry out 3 replicates at each diffusion temperature to obtain an average rate of glucose diffusion to ensure reliability of the experiment;
- Repeat the entire experiment twice to ensure the reproducibility of the results;

Control [0.5m each, max 1m]

- Repeat the entire experiment, replacing the solution in the Visking tubing with an **equal volume of distilled water**;
- No brick red precipitate produced from the solution;
- Presence of glucose in the distilled water in the boiling tube is due to the presence of glucose solution in the Visking tubing;

Results [0.5m each, max 2m]

- **Tabulation** of data with **headings and units** (temperature / °C, average glucose concentration in boiling tube solution in 2 minutes / g 100 cm³);
- Processing of data, includes average rate of glucose diffusion (g 100cm⁻³s⁻¹);
- Graph showing trend of average glucose diffusion vs. temperature;
- Correct labeling of x and y-axis with units;

Risk assessment [0.5m each, max 1m]

- Risk: Benedict's solution is an irritant to skin and the eyes;
- Precaution: Wear gloves and goggles to avoid contact with skin and eyes respectively;

Free-response question

Write your answers to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5 (a) Outline nucleic acid hybridisation and explain how it can be used in the detection of DNA sequences. [9]

- The formation of a double-stranded nucleic acid molecule by binding of one single stranded nucleic acid to another;
- Via **complementary base-pairing** / formation of hydrogen bonds between complementary bases;
- Possible hybrid nucleic acid molecules, two strands of DNA, two strands of RNA or DNA-RNA hybrid molecules;

[max 2m]

- **Southern blotting** is a method that uses nucleic acid hybridisation to detect specific DNA sequences;
- DNA fragments of different sizes separated by **gel electrophoresis**
- Double-stranded DNA are **denatured** into single strands by adding alkali to the gel;
- The denatured DNA is transferred to a piece of nitrocellulose membrane by **capillary action**;

[max 2m]

- **Colony hybridisation** is a method that uses nucleic acid hybridisation to identify which bacteria colonies on an agar plate contains a specific DNA sequence of interest / target DNA sequence;
- Colonies on an agar plate is pressed against a nitrocellulose membrane
- Treated with alkali to lyse cells and denature DNA;

[max 2m]

- membrane then **incubated with a labelled single-stranded DNA or RNA probe**;
- Probe has **complementary sequence** to the target DNA sequence / sequence of interest;
- If target DNA sequence is present, probe will hybridise with the target DNA and indicate its presence;
- Hybridised probe is detected via autoradiography, colour development or development of fluorescence on membrane;
- Nitrocellulose membrane will show a visual band where the probe hybridised with the complementary target DNA;

[max 3m]

(b) Explain the advantages and disadvantages of tissue culture over more traditional methods of cloning plants, such as taking cuttings or grafting. [5]

Advantages:

- **Less space** is required to grow the plants in the lab, as compared to the field.
- Plant cultures (stored in culture flasks) are **easier and cheaper to transport** (via air freight) compared to adult plants

- **Disease-free** plants can be produced
- **Genetic modification** of plant cells is possible during tissue culture.
- Production is **not affected by weather / environmental conditions**.
- Plants that are difficult to germinate from seeds (e.g. orchids) can be easily produced via tissue culture.

[max 2m/3m]

Disadvantages:

- **Expensive** because of labour intensive/equipment/ requirement to maintain a sterile environment via e.g. use of laminar flow hoods which are expensive.
- An infected stock plant (with fungi) can produce many infected progeny/some plants are very difficult to disinfect of fungal organisms.
- Not all plants can be successfully tissue-cultured, often because the growth medium (for that plant species) is not known.
- Monocultures / (genetically identical plants) are susceptible to new diseases or pests.

[max 2m/3m]

Accept converse if student talked about advantages/disadvantages of traditional methods of plant cloning.

- (c) Describe the goals of the Human Genome Project, and with a named example, explain how the findings of the HGP may benefit the genetic modification of crops. [6]

Goal of HGP [max 4m]	Possible benefit to plant cloning [max 2m]
• Knowledge of plant genomics: To <u>determine the sequence</u> of the coded information contained in the DNA of the various genomes studied; OR • <u>Completing the sequences of several other organisms</u> (such as bacteria, yeast, <i>Drosophila melanogaster</i> and the mouse) to be used as models for research → Knowledge gained by the study of genomes of other organisms would assist in the analysis of the human genome;	• Give example: determine sequence of - o <i>psy</i> gene (from daffodil) - o <i>crtI</i> gene (from soil bacterium); OR - o <i>Bt</i> gene (from soil bacterium/ <i>Bacillus thuringiensis</i>) • (In the process of sequencing and mapping of human genome), <u>genetic engineering/ techniques of cloning</u> were developed (which could be applied to create genetically modified crops);
• Functional genomics: <u>studying genes' normal functions</u>;	• Idea of allowing the isolation of gene(s) of interest from other organisms which code for traits which are desirable to have in GM crop; • Give example: - o Golden rice: <i>psy</i> gene and <i>crtI</i> gene code for enzymes (phytoene synthase and carotene desaturase) that convert phytoene to β-carotene (in rice endosperm) (→ improve nutritious value/increase quality of rice); OR - o Bt corn: <i>Bt</i> gene codes for toxin which causes the death of corn borer (→ increase crop yield without use of chemical insecticides/pesticides)

	to improve the yield and quality of crop plants to solve world demand for food (don't double award if benefits are explained with the examples);
Storing all sequence information in databases that are accessible by all;	Idea of: easy access of shared data enables different groups of researchers to search for information about e.g. gene sequences and functions without having to do extensive/time-consuming research on their own to create new GM crops with novel traits;
Studying and addressing the ethical, legal and social implications (ELSI) of genome research;	
Improving tools for analysis such as sequencing technology development, developing technology for functional genomics, developing technology in bioinformatics;	

[Total:20]